

The VHDL code (LCD_fpga.vhd) for implementing the LCD controller is included in this month's EFY DVD. The code writes 'WELCOME' on the LCD display. The LCD display used here is 16×2. One can use any display and change the initialisation commands accordingly.

VHDL code description

The code begins with the standard library to be included. The ENTITY section defines the interface between the LCD module and the outside world. It includes all the input and the output connections including the 50MHz clock (Clock) and reset (Reset) as inputs, LCD register select (LCD_RS) and LCD enable (LCD_E) as outputs, LCD read/write (LCD_RW) as buffer and 8-bit data bus (DATA_BUS) as inout. DATA_BUS is defined as inout so that it is capable for both input and output operations.

The architecture section defines the two operations of the LCD module for generating the LCD clock and implementing the FSM. The different signals defined in the architecture include state and next_command of state_type, DATA_BUS_VALUE which is an 8-bit STD_LOGIC_VECTOR, COUNT_CLK_LCD which is a 28-bit STD_LOGIC_VECTOR and CLK_LCD which is a STD_LOGIC signal. The value of the signal DATA_BUS_VALUE is ported to port DATA_BUS when LCD_RW is at logic '0', otherwise the port DATA_BUS remains in the high-impedance state.

The state_type is a signal type that defines different states of the LCD, namely Hold, Display_ON, Mode_SET, Write_CHAR1, Write_CHAR2, Write_CHAR3, Write_CHAR4, Write_CHAR5, Write_CHAR6, Write_CHAR7, Return_HOME, Toggle_E, Reset1, Reset2, Reset3, Display_OFF and Display_CLR. The state and next_command are two signals of state_type.

As mentioned before, the first operation generates the slow clock for the LCD. It uses the 21st bit of the COUNT_CLK_LCD signal; the clock signal for the LCD is named as CLK_LCD.

The second operation is the FSM that drives the LCD. If the Reset pin is

TABLE II
Addresses for
Different LCD Sizes

LCD type	Address
16×2	80 81 82 83 84.....8F
	C0 C1 C2 C3 C4.....CF
20×1	80 81 82 83 84.....93
	C0 C1 C2 C3 C4.....D3
20×2	80 81 82 83 84.....93
	C0 C1 C2 C3 C4.....D3
	80 81 82 83 84.....8F
	C0 C1 C2 C3 C4.....CF
40×2	94 95 96 97 98.....A7
	D4 D5 D6 D7 D8.....E7
	80 81 82 83 84.....A7
	C0 C1 C2 C3 C4.....E7

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

in high state, the LCD is initialised by sending 0×38 in hex on the data bus. RW and RS signals are set to '0' and the Enable pin is set to '1'. The current state is defined as Reset1 and the next_command state as Reset2. When the Reset pin is in low state, the FSM of driving the LCD is implemented. The different states of the FSM are Reset1, Reset2, Reset3, Display_OFF, Display_CLR, Display_ON, Mode_SET, Write_CHAR1, Write_CHAR2, Write_CHAR3, Write_CHAR4, Write_CHAR5, Write_CHAR6, Write_CHAR7 and Return_HOME. Toggle_E and Hold are two sub-states.

For LCD initialisation, the command '0×38' (hex) is sent to LCD in three states: Reset1, Reset2 and Reset3. It may be mentioned here that the command is dependent on the size of the LCD being used. The '0×38' is used to initialise a 16×2 LCD. The RS and RW signals are set at logic '0'. Enable signal is set at logic '1'. The FSM is directed to sub-state Toggle_E where the Enable signal is made to go to logic '0' and the FSM is directed to the Hold sub-state. The Hold sub-state directs the FSM to the next state next_command as set in the main states earlier. Remember that

the data on the data bus is latched by the LCD on the high-to-low transition of the Enable signal.

After the Reset3 state, the FSM goes to the Display_OFF state. The data bus sends command '0×08' (hex) to the LCD which is for display off and cursor off. The program loop then goes to the Display_CLR state. The state clears any text on the display. To do this command '0×01' (hex) is sent to the data bus. The next state is the Display_ON state (display on and cursor off) which is executed by sending command '0×0C' (hex) on the data bus. Next state is the Mode_SET state that sets the LCD such that the cursor shifts to right after each character is displayed. This is done by setting data bus to '0×06' (hex).

The next seven states Write_CHAR1, Write_CHAR2, Write_CHAR3, Write_CHAR4, Write_CHAR5, Write_CHAR6 and Write_CHAR7 display seven alphabets 'W', 'E', 'L', 'C', 'O', 'M' and 'E' on to the display. The next state is RETURN_HOME that sets the address of the next command to be written and then sends the FSM to the Reset1 state. 'WELCOME' is again written on the LCD and the process is repeated again.

It may be mentioned here that the default position of the cursor is the beginning of the first line. If any other position needs to be used to display the text, a command needs to be sent to force the cursor to the desired position. Table II gives the addresses of different positions of the LCD.

Conclusion

This part described the LCD controller implementation using an FPGA. The next part will cover details on how to establish an RS-232 communication link using an FPGA.

To be continued...

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